

# ST O5: Stochastic Models in Finance – Assignment

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## DATA

Source: NSE India Official Website

Stock: ICICI BANK

Index: NIFTY 50

Recorded dates: 20 March – 28 March (Excluding Market Holidays)

### Futures

#### 1) ICICI BANK:

Spot Price: 367.25

Expiry Date: 26 March 2020

| Time (T) | Days for Expiry | Futures Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 9               | 366.85        | 367.25                |
| Day 1    | 8               | 354.85        | 355.05                |
| Day 2    | 7               | 337.45        | 338.55                |
| Day 3    | 6               | 345.65        | 345.7                 |
| Day 4    | 3               | 282.6         | 284                   |
| Day 5    | 2               | 292.4         | 296.5                 |
| Day 6    | 1               | 315.9         | 316.9                 |
| Day 7    | 0               | 330.45        | 330.25                |

Table 1

#### 2) NIFTY 50

Spot Price: 8967.05

Expiry Date: 26 March 2020

| Time (T) | Days for Expiry | Futures Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 9               | 8915.6        | 8967.05               |
| Day 1    | 8               | 8433.7        | 8468.8                |
| Day 2    | 7               | 8205.9        | 8263.45               |
| Day 3    | 6               | 8723.1        | 8745.45               |
| Day 4    | 3               | 7581.55       | 7610.25               |
| Day 5    | 2               | 7849.9        | 7801.05               |
| Day 6    | 1               | 8362.2        | 8317.85               |
| Day 7    | 0               | 8630.3        | 8641.45               |

Table 2

## Call Option

All the options are taken as the **European Options**.

## 1) ICICI BANK:

Spot Price: 345.7

Strike Price: 350.00

Expiry Date: 30 April 2020

| Time (T) | Days for Expiry | Options Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 41              | 46.1          | 345.7                 |
| Day 1    | 38              | 19.65         | 284                   |
| Day 2    | 37              | 23.1          | 296.5                 |
| Day 3    | 36              | 27.7          | 316.9                 |
| Day 4    | 35              | 32.85         | 330.25                |
| Day 5    | 34              | 37.8          | 339.85                |
| Day 6    | 31              | 24.65         | 313.4                 |

Table 3

## 2) NIFTY 50:

Spot Price: 8745.45

Strike Price: 13500.00

Expiry Date: 30 April 2020

| Time (T) | Days for Expiry | Options Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 41              | 6.45          | 8745.45               |
| Day 1    | 38              | 7.45          | 7610.25               |
| Day 2    | 37              | 8.6           | 7801.05               |
| Day 3    | 36              | 6.05          | 8317.85               |
| Day 4    | 35              | 3.2           | 8641.45               |
| Day 5    | 34              | 4.05          | 8660.25               |
| Day 6    | 31              | 3.7           | 8281.10               |

Table 4

## Put Option

## 1) ICICI BANK:

Spot Price: 345.7

Strike Price: 350.00

Expiry Date: 30 April 2020

| Time (T) | Days for Expiry | Options Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 41              | 47.45         | 345.7                 |
| Day 1    | 38              | 47.45         | 284                   |
| Day 2    | 37              | 75            | 296.5                 |
| Day 3    | 36              | 56.7          | 316.9                 |
| Day 4    | 35              | 55.95         | 330.25                |
| Day 5    | 34              | 48.9          | 339.85                |
| Day 6    | 31              | 59.5          | 313.4                 |

Table 5

## 2) NIFTY 50:

Spot Price: 8745.45

Strike Price: 13500

Expiry Date: 30 April 2020

| Time (T) | Days for Expiry | Options Price | Underlying Value S(T) |
|----------|-----------------|---------------|-----------------------|
| Day 0    | 41              | 4667.3        | 8745.45               |
| Day 1    | 38              | 5791.4        | 7610.25               |
| Day 2    | 37              | 5506.25       | 7801.05               |
| Day 3    | 36              | 4964.6        | 8317.85               |
| Day 4    | 35              | 4670.95       | 8641.45               |
| Day 5    | 34              | 4765.1        | 8660.25               |
| Day 6    | 31              | 5143.55       | 8281.10               |

Table 6

## Assumptions, Terminologies and Notations

Throughout this assignment, we will have following assumptions:

- i) There is no or negligible transaction cost during the trading.
- ii) There is same tax rate on all trading profits.
- iii) Market participants can borrow and lend money at same risk-free rate of interest
- iv) One can always take advantage of arbitrage opportunities.
- v) Short selling is possible.

Unless specified otherwise, we will also assume that there is no income (dividend) from the underlying asset.

### Terminologies:

T: Delivery time/expiry time

S: Price of the underlying asset today

r: Risk free rate of interest = **0.0483 (4.95%)**<sup>#</sup>

F<sub>0</sub>: Futures Price

C: Price of a European Call (CE) option

P: Price of a European Put (PE) option

K: Strike Price

$\sigma$ : Volatility

All the other notations are defined as and when required.

<sup>#</sup> - RBI Interest rate on 91-Day Treasury Bill (Primary Yield) as on 20/03/2020

## Question 1

**Arbitrage opportunity:** A situation in a market where risk free profits can be made. We have to check for an 'arbitrage' opportunity for any one derivative.

### Futures:

For Future contracts, the 'arbitrage' free price should be  $F_0 = S_0 e^{rT}$ .

| $F_0 > S_0 e^{rT}$                                                                                                                                                                                                                                                                                                                                                            | $F_0 < S_0 e^{rT}$                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>At t=0:</b><br/>Borrow <math>S_0</math> at risk free rate of interest 'r' for time 'T'. Buy the asset for <math>S_0</math> and enter into futures contract for sell it for <math>F_0</math> at time 'T'.</p> <p><b>At t=T:</b><br/>Sell the asset for <math>F_0</math>, return <math>S_0 e^{rT}</math>.</p> <p><b>Profit = <math>F_0 - S_0 e^{rT} &gt; 0</math></b></p> | <p><b>At t=0:</b><br/>Short sell the asset at <math>S_0</math>, invest at risk free rate of interest 'r' for time 'T'. Enter into futures contract to buy asset at <math>F_0</math> at 'T' i.e. Take long position in the contract</p> <p><b>At t=T:</b><br/>Get <math>S_0 e^{rT}</math> from bank, buy the asset <math>F_0</math> and return it (close the position)</p> <p><b>Profit = <math>S_0 e^{rT} - F_0 &gt; 0</math></b></p> |

We can see that both the situations lead to arbitrage opportunities. Hence,  $F_0 = S_0 e^{rT}$ . By and large the fair value ( $S_0 e^{rT}$ ) reflects where the futures should be trading at a given risk free rate and number of days to expiry.

Now, let us consider our data.

Note that, if participant enters into future contract on  $n^{\text{th}}$  day, then  $S_0$  is nothing but the underlying value of the asset on that  $n^{\text{th}}$  day.

### ICICI BANK:

| Day | Futures Price ( $F_0$ ) | Value of Asset ( $S(T)$ ) | T (in yrs.) | $S_0 e^{rT}$ | $F_0 - S_0 e^{rT}$ |
|-----|-------------------------|---------------------------|-------------|--------------|--------------------|
| 0   | 366.85                  | 367.25                    | 9/365       | 367.68764    | -0.8376402         |
| 1   | 354.85                  | 355.05                    | 8/365       | 355.426066   | -0.5760657         |
| 2   | 337.45                  | 338.55                    | 7/365       | 338.863745   | -1.4137446         |
| 3   | 345.65                  | 345.7                     | 6/365       | 345.974585   | -0.3245853         |
| 4   | 282.6                   | 284                       | 3/365       | 284.112766   | -1.5127665         |
| 5   | 292.4                   | 296.5                     | 2/265       | 296.57841    | -4.1784813         |
| 6   | 315.9                   | 316.9                     | 1/265       | 316.941938   | -1.0419378         |

Table 7

From Table 7, we can see that for all the days, there is **an arbitrage opportunity exists**. But it is important to note that the difference between  $F_0$  and  $Se^{rT}$  is negligible ( $<1\%$ ) in all but one day. These differences mainly occur due to market costs such as transaction charges, taxes, margins etc. As stated earlier, we have assumed that all these costs are zero. On day 5, the difference is significant. Hence, we can say that arbitrage opportunity may exists on that day even in actual scenario.

#### NIFTY 50:

| Day | Futures Price ( $F_0$ ) | Value of Asset ( $S(T)$ ) | T (in yrs.) | $Se^{rT}$  | $F_0 - Se^{rT}$   |
|-----|-------------------------|---------------------------|-------------|------------|-------------------|
| 0   | 8915.6                  | 8967.05                   | 9/365       | 8977.73575 | <b>-62.13575</b>  |
| 1   | 8433.7                  | 8468.8                    | 8/365       | 8477.77007 | <b>-44.070074</b> |
| 2   | 8205.9                  | 8263.45                   | 7/365       | 8271.10799 | <b>-65.207991</b> |
| 3   | 8723.1                  | 8745.45                   | 6/365       | 8752.3964  | <b>-29.296405</b> |
| 4   | 7581.55                 | 7610.25                   | 3/365       | 7613.27176 | <b>-31.721765</b> |
| 5   | 7849.9                  | 7801.05                   | 2/365       | 7803.11488 | <b>46.7851201</b> |
| 6   | 8362.2                  | 8317.85                   | 1/365       | 8318.95076 | <b>43.2492363</b> |

Table 8

Again, as in previous case, the differences between  $F_0$  and  $Se^{rT}$  are less than 1% which may occur due to transaction and other taxes. Still, in our setup there is an **‘arbitrage’ opportunity every day**.

For Options also, we can check whether an ‘arbitrage’ opportunity exists or not at any given day.

## Question 2

**In the Money (ITM):** An ITM option is one with a strike price that has already been surpassed by the current stock price.

**Out of the Money (OTM):** An OTM option is one that has a strike price that the underlying security has yet to reach.

A call option gives the option buyer the right to buy shares at the strike price if it is beneficial to do so. An in the money **call option**, therefore, is one that has a **strike price lower than the current stock price**.

Put options are purchased by traders who believe the stock price will go down. **ITM put options**, therefore, are those that have **strike prices above the current stock price**.

### ICICI BANK:

| Day | Strike Price (K) | Stock Price (S) | Call option | Put Option |
|-----|------------------|-----------------|-------------|------------|
| 0   | 350.00           | 345.7           | OTM         | ITM        |
| 1   | 350.00           | 284             | OTM         | ITM        |
| 2   | 350.00           | 296.5           | OTM         | ITM        |
| 3   | 350.00           | 316.9           | OTM         | ITM        |
| 4   | 350.00           | 330.25          | OTM         | ITM        |
| 5   | 350.00           | 339.85          | OTM         | ITM        |
| 6   | 350.00           | 313.4           | OTM         | ITM        |

Table 9

### NIFTY 50:

| Day | Strike Price (K) | Index Price (S) | Call option | Put Option |
|-----|------------------|-----------------|-------------|------------|
| 0   | 13500.00         | 8745.45         | OTM         | ITM        |
| 1   | 13500.00         | 7610.25         | OTM         | ITM        |
| 2   | 13500.00         | 7801.05         | OTM         | ITM        |
| 3   | 13500.00         | 8317.85         | OTM         | ITM        |
| 4   | 13500.00         | 8641.45         | OTM         | ITM        |
| 5   | 13500.00         | 8660.25         | OTM         | ITM        |
| 6   | 13500.00         | 8281.10         | OTM         | ITM        |

Table 10

### Question 3

#### **Short or Long Position: Which is better?**

- 1) Suppose we have a future contract on **ICICI BANK** with expiry date 26 March 2020.

According to [our data](#), Following is the situation on 17 March 2020 (Day 0):

Initial Stock price ( $S_0$ ) = 367.25, Futures Price ( $F_0$ ) = 366.85

Price on Expiry Date ( $S(T)$ ) = 316.9,  $T = 9/365$  years

#### **Long Position:**

**At  $t = 0$ :** Borrow  $S_0$  from the bank at 'r' for time 'T'. Buy the asset at  $S_0$  and enter into futures contract to sell the asset for  $F_0$  at time 'T'.

**At  $t = T$ :** Sell the asset for  $F_0$ , return  $S_0e^{rT}$  to bank.

**Profit** =  $F_0 - S_0e^{rT} = 366.85 - 367.25 \times 1.001192 = -0.837641$

#### **Short Position:**

**At  $t = 0$ :** Short sell the asset at  $S_0$  and invest at risk-free interest 'r' for time 'T'. Enter into futures contract to buy it for  $F_0$  at time 'T'.

**At  $t = T$ :** Get  $S_0e^{rT}$  from the bank, buy the asset for  $F_0$  and return it (close the short position)

**Profit** =  $S_0e^{rT} - F_0 = 0.837641$

Hence, taking a **long position gives profit**.

- 2) Suppose we have a future contract on an index NIFTY 50 with expiry date 26 March 2020. According to [our data](#), Following is the situation on 17 March 2020 (Day 0):

Initial Stock price ( $S_0$ ) = 8967.05, Futures Price ( $F_0$ ) = 8915.6

Price on Expiry Date ( $S(T)$ ) = 8317.85,  $T = 9/365$  years

#### **Long Position:**

**At  $t = 0$ :** Borrow  $S_0$  from the bank at 'r' for time 'T'. Buy the asset at  $S_0$  and enter into futures contract to sell the asset for  $F_0$  at time 'T'.

**At  $t = T$ :** Sell the asset for  $F_0$ , return  $S_0e^{rT}$  to bank.

**Loss** =  $S_0e^{rT} - F_0 = 8967.05 \times 1.001192 - 8915.6 = 62.1387236$

#### **Short Position:**

**At  $t = 0$ :** Short sell the asset at  $S_0$  and invest at risk-free interest 'r' for time 'T'. Enter into futures contract to buy it for  $F_0$  at time 'T'.

**At  $t = T$ :** Get  $S_0e^{rT}$  from the bank, buy the asset for  $F_0$  and return it (close the short position)

**Profit** =  $S_0e^{rT} - F_0 = 62.1387236$

Hence, taking a **short position gives profit**.

3) Suppose now we are in Options market. For **ICICI Bank**, following are the [details](#) as on Day 0:

Spot Price ( $S_0$ ) = 345.7, Expiry date = Day 10 (Assumption),  $T = 10/365$  yrs.

Call Option Price ( $C$ ) = 46.1, Strike Price ( $K$ ) = 350, Price at Expiry ( $S(T)$ ) = 313.4

**Long Position:**

**At  $t = 0$ :** Borrow 'C' amount from the bank at 'r' for time 'T'. Buy a call option for time 'T'.

**At  $t = T$ :** As  $S(T) < K$ , call option is worthless.

$$\text{Loss} = Ce^{rT} = 46.1610438$$

**Short Position:**

**At  $t = 0$ :** Assume a role of writer and sell a call option at 'C' for time 'T'

**At  $t = T$ :** As  $S(T) < K$ , call option is worthless. Buyer will not exercise the option.

$$\text{Profit} = Ce^{rT} = 46.1610438$$

Hence, taking a **short position gives profit**.

For call option, **long call strategy would be used** if the investor was **heavily bullish** on a stock and expected the stock's price to rise significantly. While, **short call strategy** would be used if an investor was **heavily bearish** on a stock and expected the stock's price to fall significantly.

## Question 4

**Returns** on an asset are defined as  $S(\Delta t)/S_0$ , where  $S(\Delta t)$  is the value of an asset at ' $\Delta t$ '.

The value ' $\sigma$ ' is called the **volatility of the stock/asset returns** if ' $\sigma\sqrt{\Delta t}$ ' is the standard deviation of the returns in the period ' $\Delta t$ '.

### Estimation:

Volatility ' $\sigma$ ' is estimated from the historical data. Its estimate is given by,

$$\hat{\sigma} = \frac{1}{\sqrt{\tau}} \frac{1}{\sqrt{n-1}} \sqrt{\sum_{i=1}^n (U_i - \bar{U}_n)^2}$$

Where,

$\tau$  = Length of the interval in years;  $n+1$  = no of observations;  $U_i$ :  $\ln(\frac{S_i}{S_{i-1}})$ ;  $\bar{U}_n$ : A.M. of  $U_i$ 's

Now consider our data for **ICICI BANK**,

| Day | S(T)   | Returns    | $U_i$      | $(U_i - \bar{U}_n)^2$ |
|-----|--------|------------|------------|-----------------------|
| 0   | 345.7  | -          | -          | -                     |
| 1   | 284    | 0.82152155 | -0.085381  | 0.00612791            |
| 2   | 296.5  | 0.85768007 | 0.01870636 | 0.00066597            |
| 3   | 316.9  | 0.91669077 | 0.02889754 | 0.00129583            |
| 4   | 330.25 | 0.95530807 | 0.01792059 | 0.00062603            |
| 5   | 339.85 | 0.98307781 | 0.01244445 | 0.00038199            |
| 6   | 313.4  | 0.90656639 | -0.0351883 | 0.00078895            |

Table 11

$$\bar{U}_n = \sum_{i=1}^6 U_i / n = -0.0426004/6 = -0.0071001, \sqrt{\sum_{i=1}^n (U_i - \bar{U}_n)^2} = 0.09943178$$

Here,  $n = 6$  and  $\tau = 1/365$ .

$$\text{Therefore, } \hat{\sigma} = \frac{1}{\sqrt{\tau}} \frac{1}{\sqrt{n-1}} \sqrt{\sum_{i=1}^n (U_i - \bar{U}_n)^2} = 0.84954551$$

### Black-Scholes Model:

The Black Scholes call option formula is calculated by multiplying the stock price by the cumulative standard normal probability distribution function.

Its formula is given by,

$$C = S_t \Phi(d_1) - Ke^{-rt} \Phi(d_2)$$

Where,

$$d_1 = \frac{\ln \frac{S_t}{K} + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}; d_2 = d_1 - \sigma\sqrt{t} \text{ and } \Phi(.) \text{ is a CDF of } N(0,1).$$

Now, suppose we need to find call option price on [25th March \(Day 3\)](#).

Here,  $t$  = time for expiry = 36/365 years,  $K = 350$ ,  $S_t = 316.9$ ,  $\hat{\sigma} = 1.62305441$  and  $r = 0.0483$

From calculations, we get

$$d_1 = -0.05268766 \text{ and } d_2 = -0.3194911 \dots\dots\dots (1)$$

Hence,

$$C = S_t \Phi(d_1) - Ke^{-rt} \Phi(d_2) = \mathbf{21.27831}$$

From [Table 3](#), we can see that actual Call Option Price is **27.7**.

The difference between actual 'C' and computed 'C' is because of following:

- i) Black-Scholes Model assumes volatility ( $\sigma$ ) remains constant over the option's life, which is not the case because volatility varies with supply and demand.
- ii) This model assumes that there are no transaction costs or taxes (which is also our assumption) and also there are no arbitrage opportunities.

Due to the above, computed prices can deviate from the actual market prices.

## Question 5

**Beta ( $\beta$ ):** A beta coefficient can measure the volatility of an individual stock in comparison to the unsystematic risk of the entire market. In statistical terms, beta represents the slope of the line through a regression of data points from an individual stock's returns against those of the market. Its formula is given by,

$$\hat{\beta} = \frac{Cov(R, R_m)}{\sigma_m^2}$$

Where,

R: Return on a portfolio;  $R_m$  : Return on the index (Market)

### Calculating Beta only for a stock

Suppose we have one share of ICICI BANK in our portfolio.  
We want to calculate 'Beta' w.r.t. to index NIFTY 50.

| Day | ICICI BANK | NIFTY 50 | Returns (R) | Returns ( $R_m$ ) |
|-----|------------|----------|-------------|-------------------|
| 0   | 345.7      | 8745.45  | -           | -                 |
| 1   | 284        | 7610.25  | -0.1784784  | -0.1298046        |
| 2   | 296.5      | 7801.05  | 0.04401408  | 0.02507145        |
| 3   | 316.9      | 8317.85  | 0.0688027   | 0.06624749        |
| 5   | 339.85     | 8660.25  | 0.02906889  | 0.00217556        |
| 6   | 313.4      | 8281.1   | -0.0778285  | -0.0437805        |

Table 12

$$Cov(R, R_m) = 0.00557368, \sigma_m^2 = 0.00416981$$

$$\text{Hence, } \hat{\beta} = \frac{Cov(R, R_m)}{\sigma_m^2} = 1.33667667$$

This means that, our stock gives 1.34 times more profit than overall market if market goes up and 1.34 times more loss if market goes down.

### Calculating Beta for entire portfolio

Now, suppose we have six derivatives and a ICICI BANK stock in our portfolio. Hence, for calculating 'Beta' for portfolio, first we need to calculate Beta for options and futures.

**Delta of an Option:** Delta is the ratio that compares the change in the price of an asset to the corresponding change in the price of its derivative. For example, if a stock option has a delta value of 0.75, this means that if the underlying stock increases in price by Rs.1 per share, the option on it will rise by Rs. 0.75 per share, all else being equal.

Now, its formula is given by,

$$\delta = \Phi(d_1) \text{ for Call option .....(2)}$$

$$\text{And } \delta = \Phi(d_1) - 1 \text{ for Put Option.....(3)}$$

Where,

$$d_1 = \frac{\ln \frac{S_t}{K} + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma \sqrt{t}}$$

Now, from (1), we get

$$d1 = -0.05268766$$

Hence, for ICICI BANK, we get

$$\delta_{call} = 0.4789904 \text{ and } \delta_{put} = -0.5210096$$

Beta of a Call/Put option is given by,

$$\beta_{option} = \frac{S_t}{K} \delta * \beta_{stock} \quad \dots \text{See Ref [7]}$$

Here,  $\beta_{stock} = 1.33667667$ ,  $S_t = 313.4$ ,  $\delta_{call} = 0.4789904$ ,  $\delta_{put} = -0.5210096$  and  $K = 350$

By substituting,

$$\beta_{call} = 0.5733 \text{ and } \beta_{put} = -0.0728 \quad \dots \dots \dots (4)$$

**Futures Beta:** It is the beta of the underlying asset multiplied by the present value interest factor using the risk-free rate. (See Ref [6])

$$\beta_{futures} = e^{rT} * \beta_{asset}$$

Hence,  $\beta_{futures} = 1.338447 \quad \dots \dots \dots (5)$

Now, we need calculate 'beta' of derivatives on **NIFTY 50**.

By doing similar calculations as above, we get,

$$\sigma_{index} = 0.24335688, \delta_{call} = 0.4232611, \delta_{put} = -0.5767389, \beta_{call} = 0.2608,$$

$$\beta_{put} = -0.3553 \text{ and } \beta_{futures} = 1.0013$$

Note that, though we can mathematically compute Beta of an individual derivative (Option), it would not make sense as individually options don't carry, they carry corresponding to stock. Hence, instead of Beta, options should be looked upon by options Greeks such as delta, alpha, vega and theta.

Now,

$$\beta_{portfolio} = \sum \beta_i w_i$$

Where,  $\beta_i$ : Beta of  $i^{th}$  asset and  $w_i$ : Weight of  $i^{th}$  asset

Weights are calculated according to proportion of the investment in that asset w.r.t total investment.

| Type       | Index/Stock | Option/Future | Investment | Weight     | Beta     |
|------------|-------------|---------------|------------|------------|----------|
| Derivative | ICICI BANK  | Future        | 330.45     | 0.02350569 | 1.338447 |
| Derivative | ICICI BANK  | CE            | 46.1       | 0.0032792  | 0.5733   |
| Derivative | ICICI BANK  | PE            | 47.45      | 0.00337523 | -0.0728  |
| Derivative | NIFTY 50    | Future        | 8630.3     | 0.61389357 | 1.0013   |
| Derivative | NIFTY 50    | CE            | 6.45       | 0.0004588  | 0.2608   |
| Derivative | NIFTY 50    | PE            | 4667.3     | 0.33199605 | -0.3553  |
| Share      | ICICI BANK  | -             | 330.25     | 0.02349146 | 1.3367   |

Table 13

Note : For Computing 'Beta' of the portfolio common 5-day period between futures and options is considered.

From Table 13,

$$\beta_{portfolio} = \sum \beta_i w_i = 0.5613495$$

This means that, our portfolio is a '**low risk, low return**' portfolio. Our portfolio gives 0.56 times profit than overall market when market goes up and 0.56 times loss than overall market when market goes down.

### **Do having 10 shares affect the Beta?**

If we have only one stock in our portfolio, change in the quantity of shares of same stock will not change the 'beta' value. As, it only multiplies the returns by the quantity, beta remains constant. But if, our portfolio consists of shares of multiple stocks, then changing the quantity of shares of one stock would change the 'beta' of our portfolio as it would change 'weight' of that stock.

But, in our case we have six derivatives and a share in our portfolio. In such situation, increasing number of shares will increase the weight corresponding to the stock, which in turn changes the 'beta' of the portfolio.

### **Do Beta of a fixed portfolio changes over time?**

Yes, 'Beta' of a fixed portfolio can change over a period of time. Various economic, political activities along with company policies may change the market's response towards the stock of particular company or companies. This will results in change of behaviour of those stocks with respect to overall market, which in turn changes the 'beta' value of our portfolio.

## References

### **Books:**

- 1) John C. Hull, 'Options, Futures and other Derivatives', 10<sup>th</sup> edition, 2018, Pearson.

### **Websites:**

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- 2) [www.stockinvestor.com](http://www.stockinvestor.com)
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